

Market Geometry: Structure and Sensitivity

Combined Phase 16 + 17 Report

This report combines two complementary investigations built on the Phase 15 residualized discovery basis. Phase 17 decomposes the discovered market axes into prompt-visible features, answering *what* the model tracks. Phase 16 tests how context placement warps those axes, answering *how stable* the market encoding is under realistic prompt variation.

DATE	DISCOVERY PROMPTS	CONTEXT-ORDER PROMPTS	METHOD
24 March 2026	184 market-only	920 (A/B/C variants)	residualized PCA + feature decomposition + order test

MAIN READ

The model encodes two clean market summaries: a *leader axis* that points toward prominent assets with strong throughput, and a *dispersion axis* that points toward something conceptually like roster unevenness — better approximated by MAD than standard deviation, though neither fully explains the axis. These encodings are stable when context follows the market, but context placed *before* the market warps the market-boundary state by up to 0.070 cosine distance — and that warp projects onto the discovered market axes. Downstream integration absorbs most of the difference, so the strongest order effect is on *perception*, not final output.

LEADER AXIS

vol_1h_max

CV R² = 0.459

DISPERSION AXIS

pct_1h_mad

CV R² = 0.523

RISK PERCEPTION WARP

0.061

cosine gap at L42

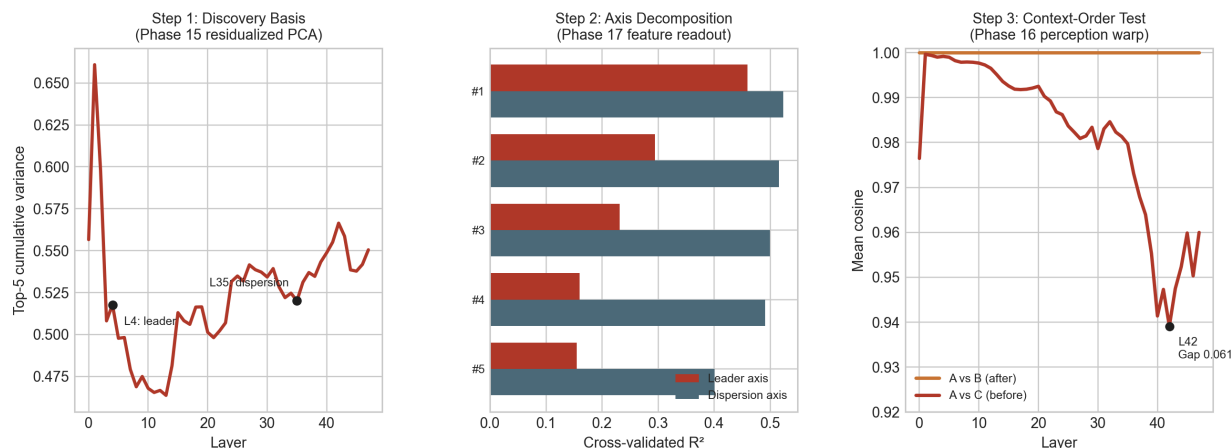
AFFORDANCE WARP

0.07

cosine gap at L40

How This Report Fits Together

This report synthesizes two phases of work that share the same foundation: the Phase 15 residualized discovery basis.



The three-step pipeline: (1) discover clean market-linked directions via residualized PCA on 184 market-only prompts, (2) decompose the strongest axes into prompt-visible features, (3) test whether context placement warps those axes.

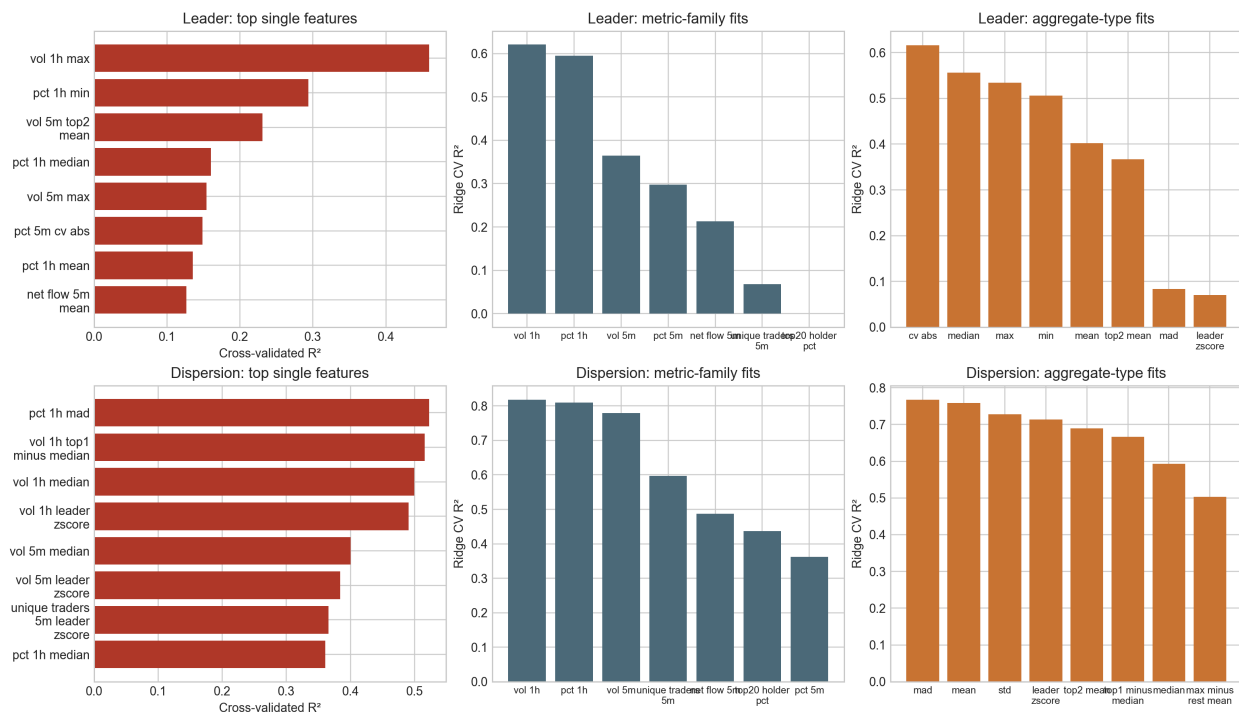
The three steps are:

1. **Discovery** (Phase 15 + rerun). Run PCA on residualized activations from 184 market-only prompts. Regress out nuisance variables (`seq_len`, `user_chars`, `n_rows`) before PCA so the top components track market content, not prompt shape. This produces a clean basis with two standout directions: an early leader axis (`market_mean`, L4, PC1) and a late dispersion axis (`market_mean`, L35, PC1).
2. **Decomposition** (Phase 17). Take those two discovered axes and ask what prompt-visible market statistics best predict the PC scores. The candidate bank contains only features computable from the rendered market rows — no hidden synthetic labels.
3. **Context-order test** (Phase 16). Take the same 184 base markets and wrap them in 920 prompt variants where risk or affordance context appears either before or after the market block. Measure whether context placement changes the market encoding at two boundary states: `market_mean` (section average) and corrected `market_eos` (last meaningful market token).

Part I: What The Market Axes Track

Phase 17 decomposes the two strongest discovered axes using only features a reader could compute from the visible market rows.

Leader Axis: market_mean, L4, PC1



Top single-feature fits, metric-family fits, and aggregate-type fits for the leader (top row) and dispersion (bottom row) axes.

The strongest single visible proxy for the leader axis is:

- vol_1h_max with CV R² = 0.459

But the family-level readout makes the interpretation richer. The vol_1h and pct_1h families dominate all others, and the best nonlinear pair combines pct_1h_max with vol_5m_max (quadratic CV R² = 0.672).

The best current interpretation: the leader axis points toward something like a *prominent asset with strong throughput* — not just the highest-return asset, but one that also has volume support. The exact nature of this axis is not fully resolved.

Dispersion Axis: market_mean, L35, PC1

The strongest single visible proxy is:

- pct_1h_mad with CV R² = 0.523

This is suggestive because the obvious candidate was pct_1h_std, which scores lower. The aggregate-type comparison points in the same direction: mad outperforms std as a proxy. But neither fully explains the axis — the best single-feature R² is still moderate, so this is a partial decomposition, not a solved readout. Conceptually, this axis appears to track something like *unevenness or leader-versus-rest structure* in the roster, though the precise formula the model uses remains open.

The best nonlinear pair is vol_5m_mean + vol_1h_median (quadratic CV R² = 0.843), which suggests the axis is multivariate: a broader spread-like market summary with co-moving volume structure, not reducible to a single statistic.

Sanity Checks

Both decompositions pass the checks that matter:

- target PC scores are nearly uncorrelated with nuisance variables after residualization
- shuffled controls collapse near zero

Axis	Best single feature	Real CV R ²	Shuffled CV R ²
Leader	vol_1h_max	0.459	< 0.01
Dispersion	pct_1h_mad	0.523	< 0.01

So the decomposition results are not being driven by prompt length, prompt size, or a probe that still works on shuffled targets.

Part II: How Context Placement Warps The Market Encoding

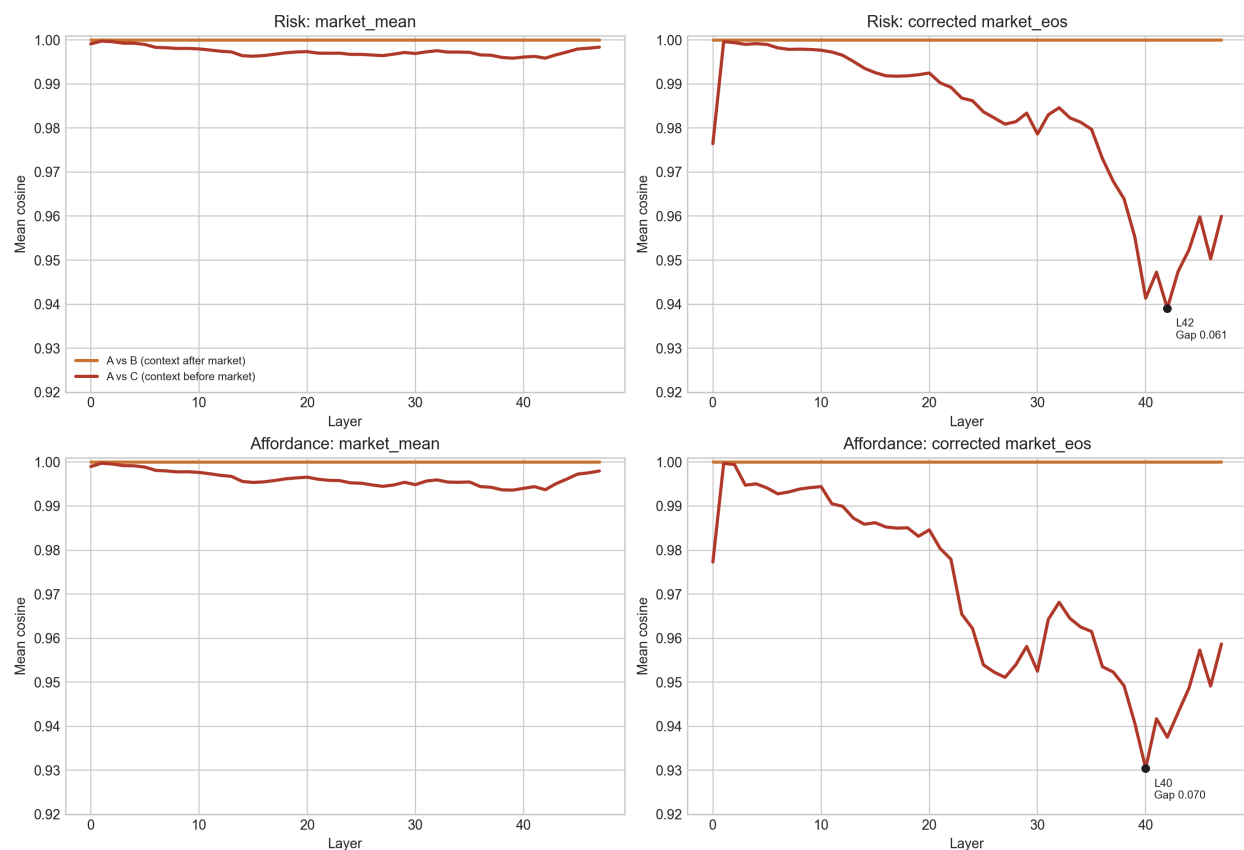
Phase 16 tests whether the discovered market axes are stable under realistic prompt variation. Each of the 184 base markets appears in five prompt variants:

Label	Prompt variant	Meaning
A	market_only	Baseline: market block first, later sections neutral.
B	context_after_market	Risk or affordance context placed <i>after</i> the market block.
C	context_before_market	Same context moved <i>before</i> the market block.

The three comparisons are:

- **Sanity check:** A vs B should be identical at market_mean and market_eos, because later context cannot retroactively change the market encoding.
- **Perception test:** A vs C measures how much earlier context warps the market read.
- **Integration test:** B vs C at downstream states measures whether the warp persists to the final output.

The Perception Warp



A vs B stays flat because later context cannot change the market encoding. A vs C pulls away once context is moved before the market, with the strongest divergence at corrected market_eos.

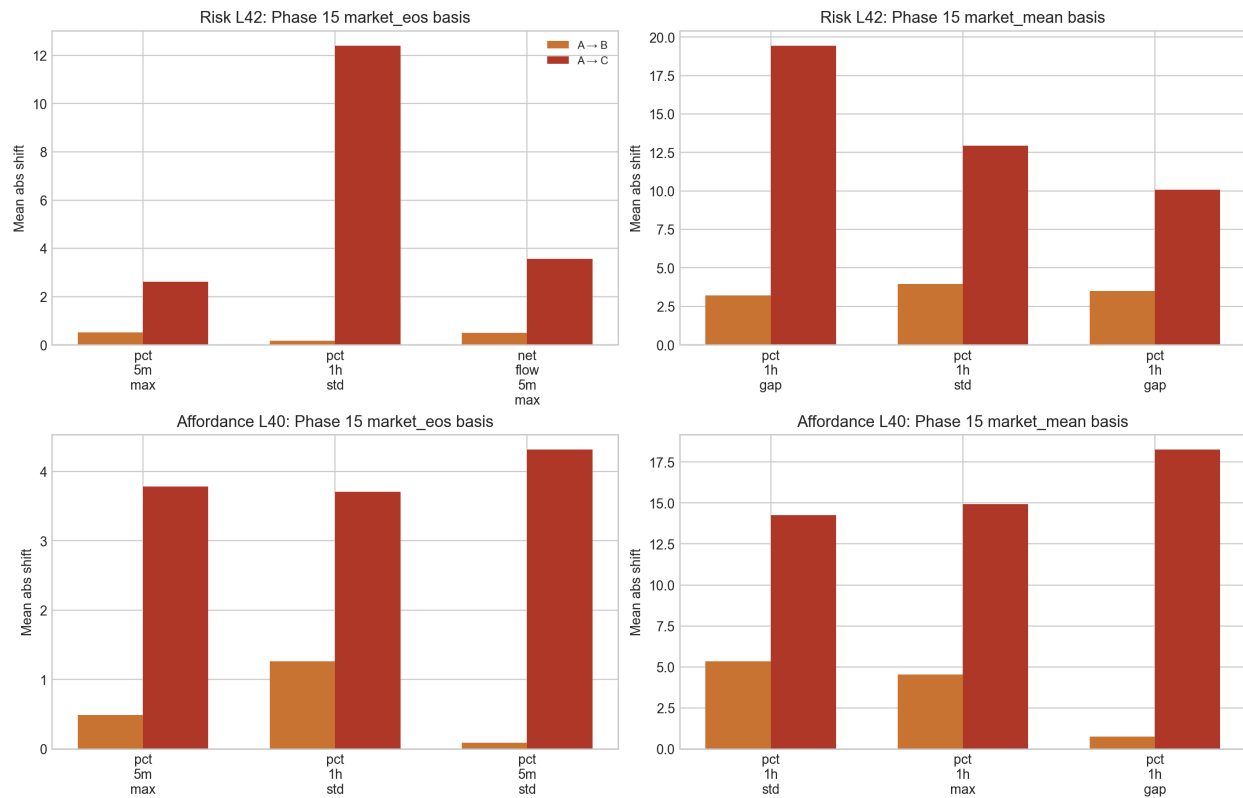
The sanity check passes exactly: A vs B is identical at both market_mean and corrected market_eos. But A vs C diverges sharply once the same context is moved before the market.

Group	State / layer	A vs B	A vs C	Gap
Risk	market_eos @ L42	1.000	0.939	0.061
Affordance	market_eos @ L40	1.000	0.930	0.070
Risk	market_mean @ L39	1.000	0.996	0.004
Affordance	market_mean @ L39	1.000	0.994	0.006

The section-average market summary is fairly robust. The precise state at the end of the market block is context-sensitive. Moving risk or affordance information before the market changes where the model *finishes* reading the market.

Where The Warp Lives In The Discovered Basis

The Phase 15 discovery basis lets us ask not just *how much* the warp is, but *where* it points.



The same corrected market_eos warp read in two Phase 15 bases. On the native market_eos basis (left), the shift is more local. On the market_mean basis (right), the same warp is re-expressed in broader roster-summary directions.

At the strongest corrected market_eos layers:

- **Risk at L42:** the A → C shift lands on PCs tied to pct_1h_std, pct_5m_max, and net_flow_5m_max
- **Affordance at L40:** the A → C shift lands on PCs tied to pct_5m_max, pct_1h_std, and pct_5m_std

When the same activations are projected onto the Phase 15 market_mean basis instead, the warp is still strong but the dominant axis labels change. For risk, the largest shift moves from pct_1h_std to pct_1h_gap. For affordance, it moves to pct_1h_gap with pct_1h_max also large.

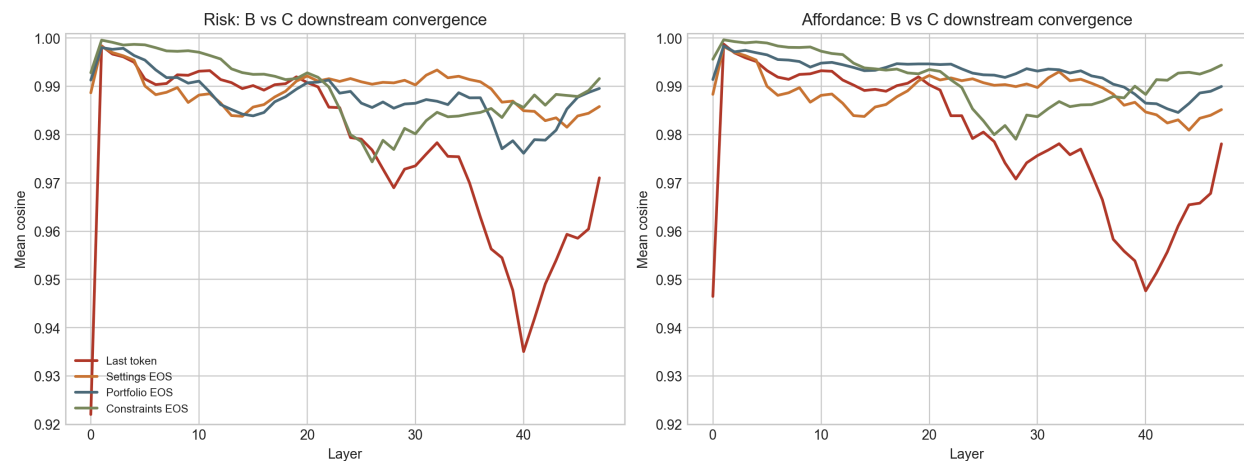
The robust conclusions:

- A → C is much larger than A → B in both bases
- the warp is real and projects onto market-linked directions

The non-robust part:

- the exact *name* of the dominant axis depends on which Phase 15 basis is used
- the basis labels are descriptive tools, not proof that the model uses one unique semantic coordinate system

Downstream Integration



Once both variants have read the full prompt, B vs C is still not identical, but much closer than the `market_eos` perception split.

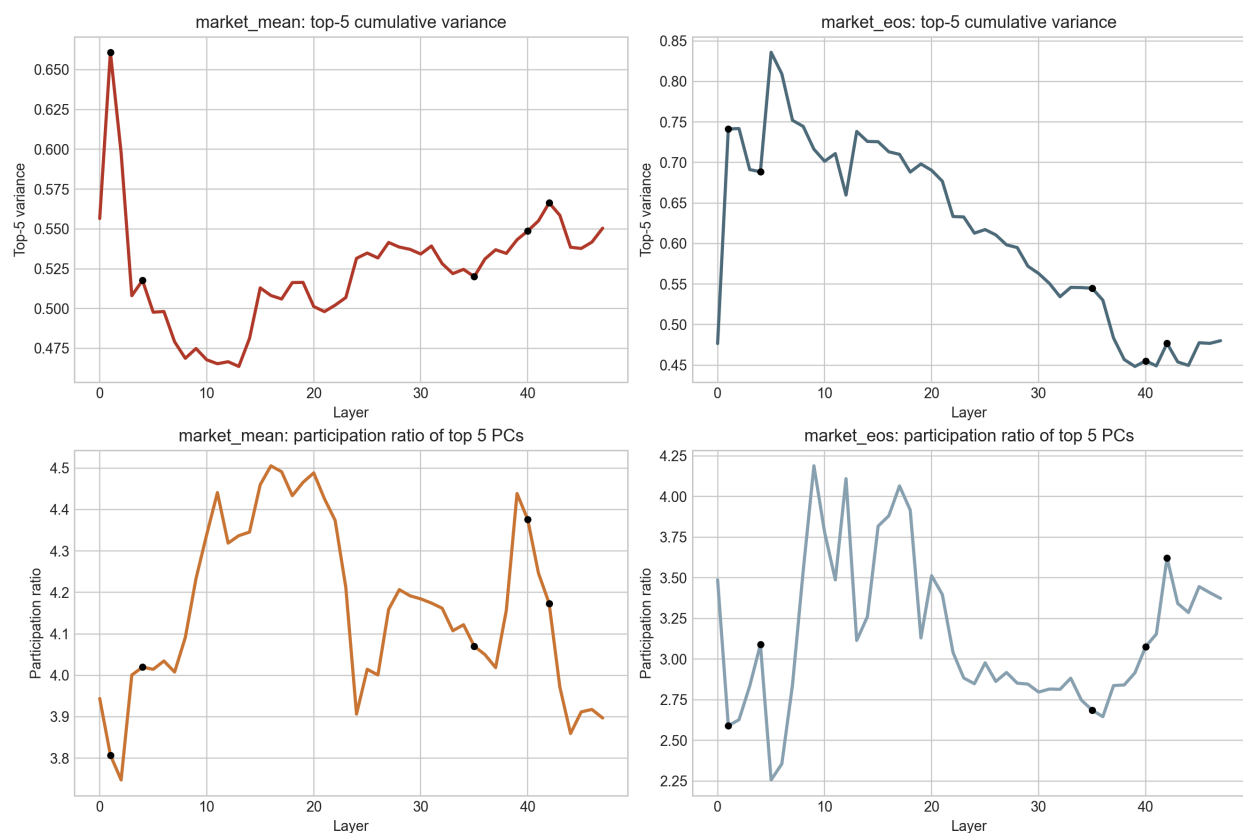
The downstream picture is forgiving:

- `last_token`, `active_settings_eos`, `portfolio_eos`, and `constraints_eos` all remain high-cosine (0.98--1.00)
- the order effect is still present but much smaller than the corrected `market_eos` split

That means context order clearly changes the market read when context is seen first, but once the model reads the whole prompt, a large part of that difference gets absorbed. The sharpest order effect is on *perception*, not final integration.

Part III: Subspace Dimensionality

The discovered market subspace is broader than one or two PCs. Within the stored top 5 PCs from Phase 15:



market_mean stays broadly distributed across 5 PCs (participation ratio ~4) while market_eos is more compressed, especially in early-to-mid layers.

State	Top-5 variance (mean)	Top-5 variance (max)	Participation ratio (mean)
market_mean	0.523	0.661	4.164
market_eos	0.613	0.836	3.15

market_mean is a broad multi-dimensional market summary, roughly ~4D in its stored top-PC readout. market_eos is more compressed but still does not collapse into one or two PCs.

Synthesis

The two phases tell a coherent story about how this model reads markets:

- 1. The model builds clean market summaries.** The discovery basis contains a leader axis (pointing toward prominent assets with volume support) and a dispersion axis (pointing toward something like roster unevenness, conceptually approximated by MAD-like measures but not fully explained by any single statistic). Both survive nuisance residualization and shuffled controls.
- 2. The summary is stable when context comes after the market.** The A vs B sanity check is exact: later context does not retroactively change the market encoding at either market_mean or corrected market_eos.

3. **Context before the market warps the market-boundary state.** The A vs C perception gap reaches 0.061 (risk) and 0.070 (affordance) at corrected market_eos. The warp projects onto the discovered market axes, not into orthogonal noise directions.
4. **The warp mostly washes out downstream.** By the time the model finishes the full prompt, B vs C reconverges to 0.98 - 1.00 cosine. The strongest order effect is on market *perception*, not the final integrated state.
5. **The subspace is broader than two axes.** The market encoding uses at least ~4 effective dimensions at market_mean. The two named axes (leader, dispersion) are the most interpretable, not the only important directions.

The practical implication: if the goal is to understand or steer the model's market read, market_mean is the better target state (cleaner, broader, more robust to context variation). The market_eos boundary is more sensitive and more informative about order effects, but also more fragile.